

Table 1: Platform engineering vs DevOps with use cases

	Platform engineering	DevOps	Use case
Goal	Developer productivity	Faster delivery	Enterprise software
Focus	Self-service platform	Build and deploy	Large teams
Users	Application developers	DevOps engineers	Multi-team organisations
Approach	Platform abstraction	Pipeline automation	Cloud platforms
Infrastructure	Platform managed	Managed manually	Kubernetes clusters
Developer work	Ready platform	Setup tools	New project
Automation	Golden paths	CI/CD pipelines	Standard workflows
Responsibility	Platform enablement	App delivery	Internal platforms
Complexity	Platform standardised	Team dependent	Large enterprises
Example	Backstage portal	Jenkins pipeline	Microservices deployment
SDR example	SDR platform	Deploy GNU Radio	Remote RF labs
AI example	ML platform	Model deployment	MLOps

Platform engineering integrates a standardised and automated high-performance development environment. It makes use of AI for optimising antenna performance and accelerating design decisions so that signals can be captured in real time without any loss.

AI-driven platform engineering is beneficial for radio frequency based communications, 6G implementations, and innovations in radio and wireless communications. The key objective of platform engineering is to speed up antenna design and simulation, and enable its automation and scaling.

The benefits of using platform engineering in radio and wireless networks include better antenna designs, consistent builds, continuous antenna signal validation, antenna design tracking, easy antenna deployment, faster setup, faster radio network testing, higher signal efficiency, optimal signal gain, reduced signal drop errors, reliable transmission results, and virtual testing.

AI and platform engineering in radio networks

The integration of artificial intelligence with platform engineering can transform RF systems when it comes to performance. Antenna and radio networks can be easily simulated, designed, and optimised for better

Table 2: Tools that can be used for platform engineering in various applications

Application	Platform engineering feature	Implementation tool
Antenna simulation	Automated pipeline	4NEC2
SDR development	Containerization	GNU Radio
RF optimization	AI integration	Python
Spectrum analysis	Shared platform	GNU Radio
Beamforming	Workflow automation	GNU Radio
Patch antenna	Version control	Git
Yagi design	Reusable templates	4NEC2
Dipole design	Parameter sweep	openEMS
RF networks	CI/CD pipeline	Jenkins
Microwave design	Standard environment	Docker
Satellite links	Cloud platform	Kubernetes
Radar systems	AI optimization	TensorFlow
IoT antennas	Automated reports	Python
MIMO systems	Resource sharing	Kubernetes
SDR testing	Automated validation	PyTest
RF prototyping	Digital twin	SimPy
EMI analysis	Batch simulation	openEMS
5G planning	Scalable platform	Docker
Antenna arrays	AI tuning	PyTorch
RF research	Shared repository	GitLab

performance in real world scenarios and wireless protocols.

AI and platform engineering can enable:

- Large scale wireless simulations on 5G/6G
- Security and integrity in existing 4G

and LTE based environments

- Wireless and radio data analysis
- Dimension recommendation and design optimization
- RF scenario analysis and prediction
- Dataset management and task automation